

Fasten a Closure Series and Fold a Garment SOP

One episode does two things. First it fastens five closures down the front of one garment. Then it folds the garment and puts it in the stack tray.

The garment is already on the fold mat when recording starts. It lies face up and spread out.

The order is always the same: zip, then button, then snap, then buckle, then velcro. You work from the collar down toward the front edge. Fasten one closure and check that it holds before you start the next one. Once a closure is fastened, you never open it again.

Nothing is folded until all five closures are fastened and the front has been smoothed flat. Nothing is folded while the garment is face up.

The right gripper does the main work. It fastens every closure, smooths the front, and folds the right side. The left gripper helps. It holds the cloth still for every closure, and it folds the left side. The two grippers work together for three things only: turning the garment over, folding it in half, and carrying the finished packet to the tray. Neither gripper does the other one's jobs.

Everything happens with the garment flat on the fold mat. Nothing is lifted into the air and worked there. Nothing is pushed or shoved into place. A gripper opens only on the fold mat, or in the stack tray when it puts the packet down.

The zip and the buckle housing are stiff. They are the only stiff parts of the garment. Never lay a fold edge across one, and never drag a gripper over one.

Setup

Complete both checklists before starting an episode.

Cell configuration

- **Environment camera:** 900 mm.
- **cell_type:** bimanual

Hardware checklist

1. The cameras are on and recording.
2. The environment camera shows the whole fold mat, the garment spread on it, and the stack tray behind it. The collar, the hem, both side edges, and all five closures are in the picture.
3. The overhead camera shows the whole front of the garment. All five closures are easy to see: the zip, the button and its buttonhole, the snap, the buckle, and the velcro.
4. Both arms are at home. Both grippers are open.

5. The desk holds nothing except the fold mat, the garment on it, and the empty stack tray.
6. The left gripper can reach the left half of the mat, the middle of the mat, the collar, the hem, and the stack tray. The arm does not have to lean out or stretch to a joint limit to get there.
7. The right gripper can reach the right half of the mat, the middle of the mat, the collar, the hem, and the stack tray. The arm does not have to lean out or stretch to a joint limit to get there.

Materials checklist

1. One **garment** lies face up and square on the fold mat. The collar points to the far side and the hem points to the front edge. No part of it hangs off a mat edge. Every episode uses the same kind of garment.
2. The garment has no sleeves. It has no ties, no hood, and no hanging strap. Both side edges run straight from the shoulder to the hem. Nothing lies across the front.
3. The front of the garment opens down the middle, so it has two front edges. The **right front edge** lies on top and carries the moving half of every closure. The **left front edge** lies underneath and carries the fixed half of every closure.
4. Five closures sit in one line down the front, well apart from each other. From the collar down they are: the **zip**, the **button**, the **snap**, the **buckle**, the **velcro**.
5. The whole line of closures, including both webbing straps of the buckle, sits in the top half of the garment's length and inside the middle third of its width. This keeps every fold edge off the closures.
6. All five closures start open. The zip slider sits at its bottom stop. The button is out of its buttonhole. The snap is apart. The buckle tongue is out of its housing. The velcro hook strap is folded back off its loop patch.
7. **Zip**: a short zip at the collar end of the front. Both **zip tapes** are already in the **slider**, so no tape ever has to be fed into it. The slider sits at the **bottom stop** and its **pull tab** lies flat, pointing toward the front edge. The slider runs all the way to the **top stop** in one smooth pull.
8. **Button**: one large flat button on the left front edge. It is wider than a gripper fingertip. It is sewn on a loose thread shank, so it stands up a little off the cloth. Its **buttonhole** is on the **button tab** on the right front edge and is clearly longer than the button is wide, so the button goes through in one straight pull.
9. **Snap**: one press snap. The **stud** is on the left front edge, facing up. The **socket** is on the right front edge, facing down. It closes with one straight press down and stays closed.
10. **Buckle**: one side release buckle. The **housing** is on a webbing strap on the left front edge. The **tongue** is on a webbing strap on the right front edge. The tongue clicks into the housing with one straight push and holds under a gentle pull. Both webbing straps lie flat and neither is twisted.

11. **Velcro:** one **hook strap** on the right front edge and one **loop patch** on the left front edge. The patch is clearly wider and longer than the strap. The strap holds on the patch under a gentle pull.
12. Every closure is undamaged and firmly attached. If a closure will not fasten, or if it opens by itself while nobody is touching the garment, fix it or swap the garment before the episode.
13. The garment is pressed flat, dry, and clean, with no stains, tears, or open seams. It holds a fold instead of springing open. It shows no fold line from an earlier episode.
14. The **fold mat** lies flat in the middle of the desk with nothing on it but the garment. It is big enough that the spread garment sits inside it with clear mat all round. Its surface grips cloth, so the garment does not slide when a gripper strokes it.
15. The **stack tray** sits at the far side of the desk, behind the fold mat. It is empty and square to the desk. It is big enough that the finished packet lies inside it with clear tray all round.

Workspace layout

- **Fold mat:** flat in the middle of the desk, where both arms can reach it. It holds one garment. Every pull, press, stroke, and fold happens here.
- **Left half of the mat:** the left side edge and the left half of the back. This is the left gripper's zone.
- **Right half of the mat:** the right side edge and the right half of the back. This is the right gripper's zone.
- **Middle of the mat:** the front, the closure line, and the collar. Both grippers can reach here, one to hold and one to work. Neither gripper carries anything across to the far side of it.
- **Stack tray:** at the far side of the desk, behind the fold mat, where both arms can reach it. It takes the finished packet and nothing else.

Vocabulary

- **Arm assignment:** which gripper does what. The right gripper fastens every closure, smooths the front, and folds the right side. The left gripper holds the cloth still for every closure, and folds the left side. The two grippers work together only to turn the garment over, to make the half fold, and to carry the packet to the tray.
- **Closure:** any one of the five fasteners down the front of the garment. A closure joins the two front edges together. This garment has five kinds: a zip, a button, a snap, a buckle, and velcro.
- **Closure line:** the five closures, sitting one below the other down the front. They are numbered from the collar down. Closure 1 is the zip, closure 2 is the button, closure 3 is the snap, closure 4 is the buckle, closure 5 is the velcro.
- **Closure order:** the order the closures are fastened in, which is always zip, then button, then snap, then buckle, then velcro. It runs from the collar down toward the front edge, and it

never changes.

- **Front:** the side of the garment that carries the closure line. When the front points at the ceiling, the garment is **face up**. When the front lies against the mat, the garment is **face down**.
- **Back:** the side of the garment opposite the front. It has no closures on it. Every fold is made with the back facing up.
- **Front edges:** the two edges that meet down the middle of the front. The **right front edge** is the one on the right side of the mat. It lies on top and carries the moving half of every closure. The **left front edge** is the one on the left side of the mat. It lies underneath and carries the fixed half of every closure.
- **Side edge:** the seam line at the side of the garment, running from the shoulder down to the hem. There is one on the left and one on the right.
- **Collar band:** the strip of cloth around the neck opening at the top of the garment. It is the part that has to stay showing in the finished packet.
- **Hem:** the finished bottom edge of the garment, at the opposite end from the collar.
- **Hem corner:** the point where the hem meets a side edge. There is a left hem corner and a right hem corner. The two grippers pinch these corners when they turn the garment over and when they fold it in half.
- **Zip:** closure 1, at the collar end of the front. It has two **zip tapes**, one sewn to each front edge, with teeth along them. A **slider** rides along the tapes and locks the teeth together behind itself. The slider is already sitting on both tapes when the episode starts.
- **Pull tab:** the small flat tag hanging off the slider. It is the only part of the zip a gripper ever holds.
- **Bottom stop:** the small block at the lower end of the zip. The slider rests against it when the zip is open. The **top stop** is the matching block at the collar end, where the slider rests when the zip is closed.
- **Closed zip:** the slider is against the top stop and the two zip tapes are locked together along the whole zip. There is no gap, and no length of tape lying loose behind the slider.
- **Button:** closure 2. It is a flat disc sewn to the left front edge on a loose thread shank, so it stands up a little off the cloth. That gap under it is what lets the buttonhole slip over it.
- **Buttonhole:** the slit in the right front edge that the button passes through. It is longer than the button is wide.
- **Button tab:** the piece of the right front edge that carries the buttonhole. This is the part a gripper pinches and pulls.
- **Buttoned:** the button is all the way through the buttonhole and stands on the outside with cloth around it on every side. The button tab lies flat and the buttonhole is not caught on the rim of the button.

- **Snap:** closure 3. It is a press fastener in two halves. The **stud** is the raised bump on the left front edge. The **socket** is the cup on the right front edge. Push the socket down onto the stud and it clicks and grips.
- **Closed snap:** the two front edges stay together at the snap after the gripper lifts off. They sit level with each other, with no gap and no step between them. Nothing less than this counts as closed.
- **Buckle:** closure 4. It is a side release buckle in two halves, each sewn to its own **webbing strap**, which is a flat woven band. The **housing** is the hollow half on the left front edge, and its opening is the **housing mouth**. The **tongue** is the pronged half on the right front edge that slides into the housing.
- **Buckled:** the tongue is pushed all the way into the housing and clicked as it went in. There is no gap between the shoulders of the tongue and the housing mouth, and it holds under a gentle pull.
- **Velcro:** closure 5, in two halves. The **hook strap** on the right front edge is the rough half. The **loop patch** on the left front edge is the soft half. Press the strap onto the patch and the two grip each other.
- **Closed velcro:** the hook strap lies flat and square on the loop patch, inside the patch edges on every side. No corner of the strap stands up, and the strap stays down when the gripper lifts off.
- **Fastened:** a closure that has reached its finished state, meaning a closed zip, a buttoned button, a closed snap, a buckled buckle, or closed velcro. A closure that only looks nearly right is not fastened.
- **Stiff part:** the zip and the buckle housing. They do not bend or flatten under a fold. No fold edge is ever laid across one and no gripper is ever dragged over one.
- **Pin:** the gripper closes, comes straight down, and presses on the cloth to hold it still. It stays there without moving until the step says it can lift. A pin does not squeeze the cloth between the fingers and it does not slide along.
- **Press:** the gripper closes, comes straight down flat on the cloth, stays still for 2 seconds, and lifts straight up. A press does not slide, rub, or pinch.
- **Pull:** the gripper holds one part and moves it to its end point in one smooth run. It does not let go partway and it does not lift the part off the cloth partway.
- **Stroke:** the gripper closes, presses down on the cloth, and runs along in one smooth line from one end to the other, staying in contact the whole way. It never lifts off in the middle.
- **Gentle pull:** the gripper holds a webbing strap or a hook strap and draws it away just far enough to show whether the closure holds. It never pulls hard enough to move the garment on the mat.
- **Lift clear:** raise the thing straight up until there is daylight under it, before moving it sideways.

- **Fold:** a gripper pinches an edge or a corner, lifts it clear of the cloth under it, carries it across, lowers it onto the spot it is going, and opens. Nothing is ever folded by pushing it along the mat.
- **Fold edge:** the closed crease that a fold makes. The other sides of a folded panel are loose open edges.
- **Half fold:** the last fold. It brings the hem edge up to the collar end. Its crease crosses the middle of the garment's length, and it stops just short of the collar band so the collar band still shows.
- **Packet:** the garment once both sides are folded in and the half fold is done. It is one squared block of layers with the whole collar band showing along the front edge.
- **Settled:** the cloth stays where it was put after the gripper lifts away, with no layer springing open.
- **Release point:** the only places a gripper is allowed to open. Those are the fold mat, and the stack tray when the packet is being put down.

Steps

Steps 1 to 11 run once, in order, on the one garment. Step 12 ends the episode.

Step 1: Close the zip

Goal: the slider sits at the top stop and the whole zip is locked shut.

- The **left gripper** closes and **pins** the **left front edge** just below the **bottom stop**. It holds there for the whole step.
- The **right gripper** pinches the **pull tab** and **pulls** the slider along the zip from the bottom stop up to the **top stop** in one smooth run. It does not let go on the way and it does not lift the slider off the cloth.
- The **right gripper** opens at the top stop.
- The left gripper lifts away only after the right gripper is off the pull tab.

Move the slider in one run. Short tugs pull the tapes out of line and can split them open behind the slider.

Check: the slider is against the top stop. The two zip tapes are locked together along the whole zip, with no gap and no loose tape behind the slider. The front edges sit level at the zip. The garment is still square on the mat.

If the slider stops partway, do not force it. Take it back down to the bottom stop and pull it up again. If the tapes split behind the slider twice, the zip is broken: stop, log it as a system issue, and discard the episode.

Step 2: Fasten the button

Goal: the button is all the way through the buttonhole and the button tab lies flat.

- The **left gripper** closes and **pins** the **left front edge** right beside the **button**. It holds there for the whole step.
- The **right gripper** pinches the **button tab** at its outer edge, past the **buttonhole**. It lifts the tab just clear of the cloth under it and lowers it so the buttonhole sits over the button.
- The **right gripper pulls** the button tab away from the button, along the front, in one smooth run. It keeps going until the far end of the buttonhole passes the rim of the button and the button stands through the hole. Then it opens.
- The **right gripper** closes and **presses** flat on the button tab beside the button, holds for 2 seconds, and lifts straight up.
- The left gripper lifts away only after the right gripper is off the cloth.

Bring the hole to the button, then pull the tab through. Pushing the buttonhole sideways at the button drags the whole front along with it, and the hole lands beside the button instead of over it.

Check: the button stands all the way through the buttonhole with cloth around it on every side. The buttonhole is not caught on the rim of the button. The button tab lies flat with no ruck in it. The zip is still closed.

If the button did not go through, lift the tab clear, lay the buttonhole over the button again, and pull again. If it still will not go through after three tries, the buttonhole is too tight: stop, log it as a system issue, and discard the episode.

Step 3: Press the snap closed

Goal: the snap is closed and the two front edges sit level at the snap.

- The **left gripper** closes and **pins** the **left front edge** right beside the **stud**. It holds there for the whole step.
- The **right gripper** pinches the **right front edge** at the **socket**, lifts it just clear of the cloth under it, lowers it so the socket sits on top of the stud, and opens.
- The **right gripper** closes and **presses** flat on the snap, holds for 2 seconds, and lifts straight up.
- The left gripper lifts away only after the right gripper is off the cloth.

Lift the right front edge and lower it onto the stud. Pulling the front edge sideways into place drags the whole front along with it, and the socket lands off the stud.

Check: the snap stays together when the gripper lifts off. The two front edges sit level at the snap, with no gap and no step between them. The zip and the button are still fastened.

If the snap did not take, lift the right front edge clear, lower it onto the stud again, and press again. If it still will not close after three presses, the snap is broken: stop, log it as a system issue, and discard the episode.

Step 4: Buckle the buckle

Goal: the tongue is seated in the housing, it clicked, and it holds under a gentle pull.

- The **left gripper** closes and **pins** the **housing** flat on the mat, with the **housing mouth** clear and facing the right front edge. It holds there for the whole step.
- The **right gripper** pinches the **webbing strap** just behind the **tongue**, lifts the tongue just clear of the cloth under it, lines it up square with the housing mouth, and pushes it straight in until it clicks. Then it opens.
- The **right gripper** takes hold of the webbing strap on the far side of the housing and gives it one **gentle pull** to show that the buckle holds. Then it opens.
- Both grippers lift away.

Line the tongue up square before pushing. A tongue pushed in at an angle jams on the lip of the housing and sits there looking seated without ever clicking.

Check: the tongue is all the way into the housing, with no gap between the shoulders of the tongue and the housing mouth. The buckle held under the gentle pull. Both webbing straps lie flat and neither is twisted. The zip, the button, and the snap are still fastened.

If the tongue did not click, or if it came out under the pull, take it out, line it up square, and push it in again. If it will not click after three pushes, the buckle is broken: stop, log it as a system issue, and discard the episode.

Step 5: Close the velcro strap

Goal: the hook strap lies square on the loop patch and is stroked down along its whole length.

- The **left gripper** closes and **pins** the **left front edge** right beside the **loop patch**. It holds there for the whole step.
- The **right gripper** pinches the free end of the **hook strap**, lifts it clear of the cloth under it, carries it over the **loop patch**, and lowers it so the strap lies square on the patch. The strap runs the same way as the patch and stays inside the patch edges on both sides. The gripper opens once the strap is resting.
- The **right gripper** closes and **strokes** along the hook strap in one smooth run, from its sewn end right out to its free end, without lifting off. This presses the whole strap down, the free end included.
- The left gripper lifts away only after the right gripper is off the cloth.

Lay the strap on, then stroke it down. A strap patted in spots holds where it was patted and lifts everywhere else.

Check: the hook strap lies square on the loop patch, with no part of it off a patch edge and no corner standing up. It stays down when the gripper lifts off. The zip, the button, the snap, and the buckle are still fastened.

If a corner stands up, or the strap landed off the patch, peel the strap back, lay it down again, and stroke it.

Step 6: Smooth the front

Goal: the front runs straight from the collar to the hem, both sides of it flat, with every closure still fastened.

- The **left gripper** closes and **pins** the front at the **collar band**. It holds there for the whole step.
- The **right gripper** closes, presses down on the right side of the front just below the collar band, and **strokes** in one smooth run down to the **hem**. It runs beside the closure line and stays clear of it.
- The **right gripper** strokes the same way down the left side of the front, again beside the closure line and clear of it.
- The **right gripper** strokes in one smooth run down the joined front edges, from just below the **velcro** to the **hem**.
- Both grippers lift away.

Keep every stroke off the closures. A gripper dragged over the zip slider or the buckle housing can catch it and pull it open.

Check: the front runs straight from the collar to the hem, with both front edges level and joined all the way below the velcro. Both sides of the front lie flat, with no ruck or pucker. All five closures are still fastened. The garment is still square on the mat with nothing off a mat edge.

If a closure came open under a stroke, fasten it again the way its own step says, then stroke that part of the front again.

Step 7: Turn the garment face down

Goal: the garment lies face down and square on the mat, with the collar toward the front edge.

- The **left gripper** pinches the **left hem corner** and the **right gripper** pinches the **right hem corner**.
- Both grippers lift the hem edge clear of the mat, carry it over the garment toward the far side, and lay it down past the collar, so the garment rolls over onto its front. They open once it is resting.
- Both grippers lift away.
- If a side edge lands folded underneath, the gripper on that side pinches it, lifts it clear, and lays it out flat again.

The garment turns over on the mat. It is never lifted off the mat and turned in the air.

Check: the garment lies face down and square, with the collar toward the front edge and the hem toward the far side. No part of it is folded underneath. No front edge gapes at the collar or the hem. All five closures are still fastened. Nothing hangs off a mat edge.

If it landed crooked or bunched, take the same two corners again and lay it down again.

Step 8: Fold the left side in

Goal: the left side is folded in on a straight fold edge that lands clear of every closure.

- The **right gripper** closes and **pins** the back in the middle of the garment. It holds there for the whole step.
- The **left gripper** pinches the **left side edge** halfway along its length, lifts it clear, carries it toward the middle, and lays it down so the new **fold edge** runs straight from the outer edge of the collar down to the hem. The fold edge runs parallel to the closure line underneath and lands clear of it. The gripper opens once the panel is resting.
- The **left gripper** closes and **strokes** along the fold edge, from the collar end to the hem end.
- Both grippers lift away.

Carry the panel over and lay it down. A panel dragged across the mat rucks up the back underneath it and lands the fold edge crooked.

Check: the fold edge runs straight from the collar to the hem, parallel to the closure line, and lands clear of the buckle and the zip underneath. There is no ruck along the fold and no part of the panel hangs past the hem edge.

If the fold edge landed crooked, landed on a closure, or came over so far that it crosses the closure line, lift that edge, lay it down again, and stroke it.

Step 9: Fold the right side in

Goal: the right side is folded in on a straight fold edge that lands clear of every closure, so the garment is now one panel with two straight sides.

- The **left gripper** closes and **pins** the back in the middle of the garment. It holds there for the whole step.
- The **right gripper** pinches the **right side edge** halfway along its length, lifts it clear, carries it toward the middle, and lays it down so the new **fold edge** runs straight from the outer edge of the collar down to the hem. The fold edge runs parallel to the left fold edge and lands clear of the closure line. The gripper opens once the panel is resting.
- The **right gripper** closes and **strokes** along the fold edge, from the collar end to the hem end.

- Both grippers lift away.

Check: the two fold edges run parallel, and each one is the same distance in from the collar as the other. Both land clear of the buckle and the zip. Nothing shows past either fold edge. The garment is now about as wide as the collar.

If the two sides came out uneven, lift the right edge, lay it down again to match the left, and stroke it.

Step 10: Fold the garment in half

Goal: the hem edge is brought up to the collar end on one straight fold edge, with the whole collar band left showing.

- The **left gripper** pinches the **left hem corner** and the **right gripper** pinches the **right hem corner**.
- Both grippers lift the hem edge clear, carry it toward the front edge, and lay it down **just short of the collar band**, so the whole collar band stays showing. The hem edge lands square to the two side fold edges. They open once it is resting.
- The **right gripper** closes and **strokes** along the new fold edge, from its left end to its right end.
- Both grippers lift away.

The crease of the **half fold** crosses the middle of the garment's length, below the closure line, so it lands on plain cloth. Look at where the buckle is before laying the edge down. A crease pressed over the buckle housing or the zip makes the packet rock, and it can pull the closure open.

Check: the garment is one squared **packet**. The fold edge runs parallel to the collar and crosses the middle of the length, clear of the buckle and the zip. The whole collar band shows along the front edge with nothing lying on it. Nothing shows past a side fold edge. No layer has sprung open.

If the fold landed crooked, covered the collar band, or creased over a closure, lift that fold edge back up, lay it down again, and stroke it.

Step 11: Put the folded garment in the stack tray

Goal: the packet lies flat and square inside the empty stack tray.

- The **left gripper** pinches the packet at its **left side fold edge** and the **right gripper** pinches it at its **right side fold edge**, each one halfway along its length.
- Both grippers lift the packet level off the mat, carry it level to the **stack tray**, and lower it until it lies flat inside the tray, square to the tray edges, with the collar band toward the front edge. They open together.
- Both grippers lift away.

Both grippers carry the packet and both open together. A packet carried on one gripper hangs from that one hold, and the layers slide apart in the air.

Check: the packet lies flat inside the stack tray, square to the tray edges. The whole collar band shows along the front edge. No part of the packet hangs over a tray edge. No layer has sprung open. The fold mat is bare.

If it landed crooked, or a layer opened, take the same two side fold edges, lift it clear, and lay it down again.

Step 12: Check the packet and end the episode

- Check that the stack tray holds one squared packet lying flat and square, that the whole collar band is showing along the front edge, that no layer has sprung open, that no front edge gapes anywhere along the packet, and that the fold mat is bare.
- Return both arms home, clear of the fold mat and the stack tray, then stop recording.

After the episode: reset the workspace

This reset is not recorded.

1. Take the folded garment out of the stack tray and open it out flat.
2. Open the five closures in the opposite order. Peel the hook strap off its loop patch. Press the buckle release and pull the tongue out of the housing. Pull the two front edges apart at the snap. Pull the button back out through its buttonhole. Run the zip slider back down to its bottom stop.
3. Check every closure. The zip runs freely and both tapes stay in the slider. The button and its thread are firm, and the buttonhole is not torn or stretched. The snap stud and socket are undamaged and still click closed under a straight press. The buckle clicks and holds under a gentle pull, with no crack in the housing. The velcro hooks and the loop patch still hold under a gentle pull. Fix or replace anything that fails.
4. Press the garment flat again, so no fold line from the last episode is left in the cloth. A garment that still shows an old fold is not staged again.
5. Replace a garment that is torn, stained, damp, or has an open seam. Use a different garment from time to time rather than running the same one every episode.
6. Lay the garment back face up and square on the fold mat, collar toward the far side and hem toward the front edge, both side edges straight, and all five closures open with the right front edge resting on the left front edge.
7. Smooth the fold mat flat, check that it has not moved, and leave nothing on it but the garment.
8. Check that the stack tray is empty and square to the desk at the far side.
9. Wipe the desk.

10. Run both Setup checklists again.

SOP violations

These are actions that break the SOP and are reviewed side by side in the review tool.

How to record a violation in review

For each violation, record the **start timestamp**, **violation name**, and **SOP rule broken**. The visible cue is what the reviewer sees. The coaching note is for retraining and is not an annotation label.

Episode handling

Tag every violation with its timestamp and name. Keep the episode with the violation tag. Do not delete it just because a rule was broken.

Violations

Violation: Closures worked out of order

- **Visible cue:** a closure lower down the line is fastened while one nearer the collar is still open, or the line is worked from the hem upward.
- **SOP rule broken:** Steps 1 to 5, the closures are fastened from the collar down in the order zip, button, snap, buckle, velcro.
- **Coaching note:** start at the collar and work down. Zip, button, snap, buckle, velcro.

Violation: Moved on from a closure that is not fastened

- **Visible cue:** a closure that did not take is left as it is and the episode moves on to the next closure or to the smoothing strokes. The slider is short of the top stop, the button sits beside the hole instead of through it, the snap gapes, the tongue is only part way into the housing, or a velcro corner stands up.
- **SOP rule broken:** Steps 1 to 5, each closure is fastened and holds before the next one is started, and Step 6 begins only once all five are fastened.
- **Coaching note:** watch it as the gripper comes off. If it does not hold, work it again before you move on.

Violation: Zip pulled in tugs or forced

- **Visible cue:** the slider is moved in short tugs with releases in between, the pull tab is let go part way along, the slider is lifted off the cloth halfway, or a stuck slider is forced on instead of being taken back to the bottom stop.
- **SOP rule broken:** Step 1, the slider moves from the bottom stop to the top stop in one smooth run, and a slider that stops partway goes back to the bottom stop first.

- **Coaching note:** one run, all the way up, without letting go. If it sticks, take it back down and go again.

Violation: Front edge dragged into place

- **Visible cue:** the button tab, the right front edge at the snap, or the hook strap is pushed, pulled, or shoved sideways to get it onto its target, instead of being pinched, lifted clear, and lowered onto it.
- **SOP rule broken:** Steps 2, 3, and 5, the moving half is lifted clear of the cloth under it and lowered onto the fixed half.
- **Coaching note:** pick it up and put it down. Dragging takes the whole front with it.

Violation: Snap or button closed by pinching or sliding

- **Visible cue:** the gripper squeezes the two layers between its fingers at the snap or the button, presses at an angle, or rubs, dabs, or slides across instead of coming down flat and holding still.
- **SOP rule broken:** Steps 2 and 3, one flat press held for 2 seconds and lifted straight up, never pinched and never slid.
- **Coaching note:** down flat, count two, straight up.

Violation: Buckle not lined up square

- **Visible cue:** the tongue goes into the housing at an angle, is worked in with repeated pushes, or jams on the lip of the housing. Or the housing is dragged toward the tongue instead of being held still. Or the gentle pull is skipped, so a tongue that never clicked is treated as buckled.
- **SOP rule broken:** Step 4, the tongue is lined up square with the housing mouth, pushed straight in until it clicks, then checked with one gentle pull.
- **Coaching note:** square it up, one straight push, then tug it once to prove it.

Violation: Velcro patted on

- **Visible cue:** the hook strap is tapped in spots, pressed only in the middle, or laid on with no stroke along it. Or it lands crooked, hangs off a patch edge, or is left with a corner standing up.
- **SOP rule broken:** Step 5, the strap lands square inside the patch edges and is stroked from its sewn end out to its free end in one smooth run.
- **Coaching note:** square it on the patch, then one run all the way out to the tip.

Violation: Closure worked without a pin

- **Visible cue:** the left gripper is off the cloth while a closure is being worked, or it only comes down after the work has started, or it lifts away before the right gripper is clear, and the front shifts while the work is happening.

- **SOP rule broken:** Steps 1 to 6, the left gripper pins the left front edge or the housing beside the closure and holds until the right gripper is off the cloth.
- **Coaching note:** pin first, let go last.

Violation: Stroke skipped, dabbed, or run over a closure

- **Visible cue:** one side of the front, the joined front edges, a side fold edge, or the half fold edge is left with no stroke along it. Or the gripper taps in spots or presses once in the middle instead of running end to end. Or a stroke is dragged across the zip slider or the buckle housing.
- **SOP rule broken:** Steps 5, 6, 8, 9, and 10, one smooth stroke end to end without lifting off, and every stroke stays off the zip and the buckle.
- **Coaching note:** one run, end to end, and go beside the closures rather than over them.

Violation: Arms swapped roles

- **Visible cue:** the left gripper works a closure, smooths the front, or folds the right side. Or the right gripper folds the left side. Both grippers taking a corner each in Steps 7, 10, and 11 is not this violation.
- **SOP rule broken:** Steps 1 to 11, the right gripper works the closures, smooths, and folds the right side, and the left gripper pins and folds the left side.
- **Coaching note:** left holds and takes the left. Right works and takes the right.

Violation: Folded before the closures were done or while still face up

- **Visible cue:** a side or the hem is folded while a closure is still open, before the front has been smoothed, or before the garment has been turned over, so a fold is made with the closure line still showing.
- **SOP rule broken:** Steps 6 and 7, all five closures are fastened and the front is smoothed before anything is folded, and the garment is turned face down first.
- **Coaching note:** fasten all five, smooth, turn. Nothing folds until the front is against the mat.

Violation: Garment turned in the air

- **Visible cue:** the garment is lifted off the mat and flipped, swung, or shaken over onto its front, instead of being carried over on the mat by its two hem corners and laid down.
- **SOP rule broken:** Step 7, the garment turns over on the mat and is never lifted off it and turned in the air.
- **Coaching note:** the wrist does not have the range to turn it. Walk it over on the mat.

Violation: Fold made by dragging

- **Visible cue:** a side edge or the hem edge is pushed, shoved, or slid across the mat into place, instead of being pinched, lifted clear, carried over, and lowered onto its spot.

- **SOP rule broken:** Steps 8, 9, and 10, the edge is lifted clear before it is carried.
- **Coaching note:** pick it up and put it down. Nothing gets pushed.

Violation: Side fold edge crooked

- **Visible cue:** a side fold edge runs crooked instead of straight from the collar to the hem, or it is not parallel to the other side, or the panel comes over so far that it crosses the closure line, or so little that the garment is left much wider than the collar.
- **SOP rule broken:** Steps 8 and 9, the fold edge runs straight from the outer edge of the collar to the hem, parallel to the closure line and to the other fold edge.
- **Coaching note:** line the edge up on the collar before you lay it down.

Violation: Fold edge laid across the zip or the buckle

- **Visible cue:** a side fold edge or the half fold edge is creased over the buckle housing or the zip, so the packet sits proud or rocks at that spot, or a closure comes open under the fold.
- **SOP rule broken:** Steps 8, 9, and 10, every fold edge lands clear of the buckle and the zip.
- **Coaching note:** find the buckle before you lay the edge down. Fold on cloth, never on hardware.

Violation: Half fold uneven or crooked

- **Visible cue:** the hem edge lands past the collar band and covers it, or lands well short of it, or the fold edge runs at an angle to the collar instead of parallel to it, or the sides of the finished packet do not line up with each other.
- **SOP rule broken:** Step 10, the hem edge lands just short of the collar band, square to the two side fold edges.
- **Coaching note:** take both corners together, keep the edge square as it comes over, and stop short of the collar band.

Violation: Packet carried or put down wrong

- **Visible cue:** the packet is carried to the stack tray on one gripper, or tilted so the layers slide, or released above the tray so it drops the last part of the way, or it lands crooked in the tray, hangs over a tray edge, or is left on the fold mat at the end of the episode.
- **SOP rule broken:** Step 11, both grippers take a side fold edge each, carry the packet level, lower it flat inside the tray square to the tray edges, and open together.
- **Coaching note:** one hold each side, keep it level, take it all the way down, then let go together.

Violation: Grip changed in the air

- **Visible cue:** a gripper shifts, rolls, or takes its hold again on the garment, a panel, a strap, or the packet without laying it down first.

- **SOP rule broken:** Steps 1 to 11, a new hold is taken only after the cloth is resting.
- **Coaching note:** put it down before you take a new hold.

Violation: Gripper opened over the wrong place

- **Visible cue:** a gripper opens above the mat or the tray and the cloth drops the last part of the way, or it opens over bare desk, over open space, or off a mat edge.
- **SOP rule broken:** Steps 1 to 11, the gripper opens on the fold mat, or in the stack tray for the packet, once the cloth is resting, and nowhere else.
- **Coaching note:** take it all the way down before you open the gripper.

Violation: Placement fixed by shoving

- **Visible cue:** a gripper nudges, pats, brushes, or straightens a panel, a strap, or the packet after letting go of it, instead of lifting that edge clear and laying it down again.
- **SOP rule broken:** Steps 5 to 11, a bad placement or fold is fixed by lifting that edge and laying it down again.
- **Coaching note:** lift it and lay it again, or leave it alone.

Violation: Closure opened again after it was fastened

- **Visible cue:** a closure that was fastened earlier in the episode is opened again, peeled back, pulled back, or released. Redoing a closure that failed its own Check is not this violation.
- **SOP rule broken:** Steps 1 to 5, a closure that is fastened stays fastened for the rest of the episode.
- **Coaching note:** once it holds, leave it. Only a closure that failed its Check gets worked again.

Violation: Garment dropped, knocked, or pushed off the mat

- **Visible cue:** the garment or the packet falls off the mat or the tray, the mat or the tray is knocked and moves, a finished fold comes apart on the way, or the two arms hit each other.
- **SOP rule broken:** Steps 1 to 11, the garment stays on the mat on a clear path, and a gripper opens only once the cloth has settled.
- **Coaching note:** check the path and the landing spot before you move.

Violation: Wrong episode ending

- **Visible cue:** the episode ends with a closure open, a layer sprung open, the collar band covered, the packet crooked or hanging over the tray, the garment still spread on the mat, or an arm away from home.
- **SOP rule broken:** Step 12, check the squared packet, the five fastened closures, and the collar band showing, return both arms home, then stop recording.

- **Coaching note:** check first. Homing is the last thing the arms do.

Failures that are not violations

These failures are not caused by how the task was run. Log them as system issues, discard the episode, and never use them for coaching.

- **Recording stopped or paused during the episode** (recording system).
- **Camera dropped frames or lost its feed** (capture system).
- **Hardware fault on an arm:** gripper failure, drift, controller caused collision, or motor error.
- **Broken zip:** it jams under a correct pull, its tapes split behind the slider twice, or the slider comes off the tapes. Fix it or swap the garment before the next episode.
- **Buttonhole too tight or button too loose:** the button will not go through after three correct pulls, or the button pulls away from the cloth. Fix it or swap the garment before the next episode.
- **Broken snap:** it will not click closed after three correct presses, it pops open by itself while nobody is touching the garment, or a stud or socket is pulling away from the cloth.
- **Broken buckle:** it will not click after three correct square pushes, it releases by itself, its housing is cracked, or a webbing strap is pulling away from the cloth.
- **Worn velcro:** the hooks or the loop patch no longer hold under a gentle pull, or the patch is pulling away from the cloth.
- **Garment will not hold a fold:** it is stroked end to end and the fold still springs open every time. Press the garment before the next episode.
- **Faulty garment:** found torn, stained, damp, with an open seam, or with a front that will not lie flat.
- **Garment too big for the mat:** it cannot lie spread inside the mat with clear mat all round, or a hem corner sits outside an arm's reach once it is staged.
- **Packet too big for the stack tray:** it cannot lie inside the tray without hanging over an edge.
- **Fold mat or stack tray moves out of place** during the episode.

Annotation subtasks (from SOP)

1. Close the zip
2. Fasten the button through its buttonhole
3. Press the snap closed
4. Buckle the buckle
5. Close the velcro strap
6. Smooth the front from the collar to the hem
7. Turn the garment face down on the mat

8. Fold one side of the garment in
9. Fold the garment in half
10. Put the folded garment in the stack tray
11. Return both arms home and end the episode